Big Data for Institutional Planning, Decision Support and academic excellence

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***Abstract*— the impact of big data although being very insubstantial in its inception however has gradually originated to make significant contributions to variety of spheres of life. Development of smart cities, big data in e-business, big data in health care sector, big data in environmental sciences and implementation of Big Data in other areas has indisputably revealed to critics what can be achieved by Big Data. Big Data undeniably is a paradigm shift in the perception of ways of understanding and studying the world. Big data can be utilized to analyze fine grained data for public and private use. Big data utilization in education sector is relatively less but it has started growing gradually. As a matter of fact as Big Data can be effectively utilized to improve education delivery, it can equally be utilized to provide decision support to higher management to attain excellence in the field of education. This research paper is a mere effort to explore various possibilities of Big Data utilization in educational establishments and their implications**.

Keywords—Big Data in Education; Decision support with Big Data; Big Data Analytics; Big Data in higher education.

I. INTRODUCTION

Data analytics and data mining have yielded some unprecedented benefits in various spheres of life in the past. In addition of yielding extraordinary advantages, these technologies provided an idea that future analytics will open up new possibilities illuminating the human intelligence, knowledge and perception. Moreover it also provided indications of the volume, variety, velocity and veracity of data (Big Data) on which analysis will be done in future. Big Data a mysterious term only a few years back has created a significant impact in various fields in the recent past. Michael Cox and David Ellsworth publish; Visualization offers an exciting challenge for computer systems: data sets are mostly relatively huge, challenging the capacities of main memory, local disk, and even remote disk. We call this the problem of *big data*. When data sets do not fit in main memory, or when they do not fit even on local disk, the most common solution is to acquire more resources [1]. It was first article in ACM digital library to use the term Big Data. Big Data in the recent past has not only been discussed a lot however Big Data utilization and implementations in diverse sectors has become very common. The concept and utilization of Big Data scopes creation of Smart Cities, smartmalls, smart parking, healthcare, sports, science and research, trading, manufacturing, retail, retail banking, media and although a bit less however in education [2] [3] [4]. Big data delivers an arrangement for transparency in manufacturing industry, which is the capability to untangle doubts such as unreliable component functioning, readiness and availability. Predictive manufacturing as an applicable approach for least down time and comprehensibility requires vast amount of data and advanced prediction tools for systematically processing the data into useful information [5]. Similarly in the many other areas Big Data implementation can do wanders like in technology "ebay uses two data warehouses at 7.5 and 40PB as well as a 40PB Hadoop cluster for search, consumer recommendations,and merchandising", "Amazon.com handles millions of back-end operations every day, as well as queries from more than half a million third-party sellers", "Facebook handles 50 billion photos from its user base", "Google was handling roughly 100 billion searches per month as of 2012" [6] [7]. This reveales the scope of implementation for Big Data and benefits it may yield to companies and consumers. Although utilization and implementation of Big Data in education sector has been abysmally low however scope and possibilities are surely available to enhance teaching and learning as well as to provide an unprecedented visualization to management for effective decision making, prevention and premption, e.g it is possible to identify who at the time of registration is likely to drop a module using learner analytics and educational data mining techniques. Big Data analytics and mining can be utilized also to provide courses and modules to the students in the form of menus hence providing menu of learning options for the students, similar to what walmart is providing to its consumers a collection of items and schemes based on data analysis like customers who took baby diapers also took baby food. Similary options can be provided to students like 'students who took course you took also took' or the students who have done well in these courses did well in these also. Big Data analytics can be certainly utilized in education sector to identify the failing students or failing teachers and the possible reasons for the failure. Moreover future course of action and institutional planning can be done to prevent such occurences if possible. Big Data will enable us

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to track almost everything small or big a student has done in the institution or a faculty member has done during the tenure of his/her stay in the institution including the major and minor contributions. The kind of visualization Big Data can provide to enhance teaching and learning and provide a decision support system for educational establishments to attain excellence in education is unprecedented. Big Data in education can also be implemented collecting huge amount of data from students using sensing devices, data analytics and predictive analytics can be performed on the data to reveal the unseen and prevent unforseen. One of the approaches in this category is to calculate the attentiveness of the students in a classroom by brain-computer interfaces (BCI) using electroencephalography (EEG). Student's attentiveness in the classroom significantly influences their learning outcomes, teaching and learning can be enhanced if any means of realizing the student's attentiveness can be attained. This data will be utmost important for "student success centers" of educational establishments to analyze individual students and in most of the cases to reach to the bottom of underlying facts which hinder student's performance. A wearable device can be used by the students which will be responsible for generating Big Data that can be analysed using data analytical tools. Data analysis and predictive analysis can be performed on the data to generate output which will be beneficial for students, teachers, administrators and executives of an educational establishment.

II. Related work

The use and implementation of Big Data in education although less in the past however many Big Data projects have successfully been implemented in education. Big Data can be utilized by educationist to decision making processes. Recent findings have revealed that big data acquisition and analytics can support educationists and the institutes in improved decision making. According to McKinsey [8], Big Data is defined as what firms cannot handle with typical database software. New and different technologies are needed to handle the 3 Vs of volume (from gigabytes to petabytes), velocity (from batch to near-time data and real-time streams), and variety (from structured records to semi structured and unstructured text [9]. The Necessity to store Big Data and analyzing huge amount of sensor-generated and user-generated data for enhanced decision making is reported in many areas as well as in education. Jyotsna [10] has discussed how horizon of education is expanding electronically. Various educational institutions are developing electronic learning modules, books, quizzes etc., to enhance understanding of concepts amongst students. They also provide assessment of students in systematic, real-time ways. Educational institutions are generating huge volumes of data, from grades or test scores to admissions or enrollment numbers while doing online evaluations and admissions respectively. In the recent research studies on data analytics in the field of education, the focus has been on terms like Educational data mining (also referred to as "EDM"/ "LA" i.e. Learning Analytics) [11].
EDM can be used to understand the learning behavior of the students as suggested by (Kulkarni, S., Rampure, G., Yadav, B., 2012). There are wide varieties of current methods popular within educational data mining like prediction/classification, clustering, association mining, and discovery with models etc. Big data analytics (BDA) is the process of investigative and observing the huge amount of different data types, to determine hidden patterns and measurable outcome for business impact. Hence, utilizing the big data analytics in the field of education will provide beneficial insights to improve the student knowledge and their academic performance. Big data analytics will play a major role in the education sector. The proposed use of EEG sensors for detecting the attention levels of students in the class room is not new. Many devices can be used for such a purpose; ROB3115 created by Conor Russomanno is one of the devices which can be used. Similarly NeuroSky Mindset is one more devices that can be used for the purpose, NeuroSky Mindset is an audio 3 headset equipped with a single, dry EEG sensor. "NeuroSky can be used as EEG sensors in the classroom so that teachers can see if the students are paying any attention to the material given to them [12].

III. Big Data for educational establishments as a decision support

Big Data can be utilized to provide unprecedented visualization to the higher administration of educational establishments which can facilitate easy decision making and positive results. Although educational establishments all over the world are implementing various information management systems to automate operations related to registration, administration and management of the institution however instead of multiple information systems being available in various educational establishments all over the world, higher administration is not getting visualization of the facts which would make them wiser and facilitate the easy decision making and organizational planning. Moreover other stake holders of the educational institutions are also not provided with a supportive or assistive system interface, like there is no provision for students so that an assistive environment can be provided to them to easily choose courses where they are most likely to get success. Big Data implementation for educational institutions will ensure that a huge amount of data that is critical to the visualization and facilitation to various stake holders is collected from various sources like documents, emails, audit logs, CCTV footages, biometric devices, online surveys, student response systems, student and faculty feedbacks and Questionnaires. Most of the unstructured huge amount of data in educational establishments remains unutilized as there is no appropriate mechanism or technology existing in educational institutions to convert this unstructured data in to useful information which will have a huge impact on the visualization available to higher management directly effecting preventive, preemptive capabilities of educational establishments giving insights of the hidden information

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which is very critical institutional success. Big Data provides data analytics and predictive analytics feature which uncovers lot of precious information critical for the success and development of an educational establishment. Many tools that can be used for data analysis in educational establishments are Qubole, Tableau, Qlik, MicroStrategy, and TIBCO Jaspersoft. These tools can be efficiently used to provide data analysis features to educational establishment where data can be centrally analyzed in one spot. Various parallel processing database engines like Vertica as well as relational database engines such as Microsoft SQL Server and the MySQL open source database, and NoSQL databases such as MongoDB can be utilized for data storage and access. Hadoop can be used as framework for the implementation of Big Data, "Hadoop is an open-source software framework for storing data and running applications on clusters of commodity hardware. It provides massive storage for any kind of data, enormous processing power and the ability to handle virtually limitless concurrent tasks or jobs"[13]. Below given figure 1.1 depicts the architecture of Big Data in educational establishments for Decision support using Hadoop.

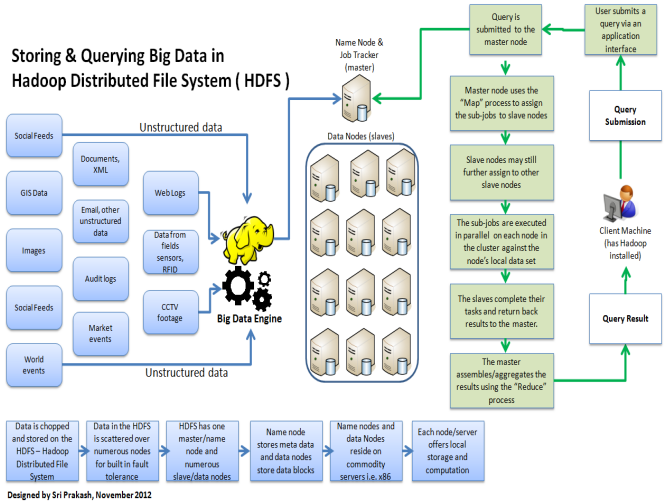

Fig. 1. Hadoop Distributed File System Sources

The above provided figure shows various sources which provide data to the Hadoop distributed file system (HDFS), the sources depicted in the figure generate unstructured data which can be stored and queried using HDFS. The sources in an educational establishment which generate unstructured but useful data on which data and predictive analysis can be conducted are several like data from feedback and survey which can be used to enhance the delivery and services in educational establishments and at the same can be used to address the grievance and future expectations of students and faculty members and planning to minimize the grievances from students as well as faculty. It can provide the trend and the areas which have resulted in most of the grievances from students as well as from faculty members. Biometric devices can provide trends of faculty members for their arrivals and departures from the institutions and correlations with their performance if any. There are various possibilities available depending upon the priorities of the educational establishments, which particular set of information, data analysis and predictive analysis they are interested in specifically. The system can be customized to provide specific visualizations which are of utmost benefit to the educational establishment. There is a lot of scope for the system to explore more and more analytics to provide maximum visualization to various stake holders in educational establishments. Figure 1.1 shows how a client who can be any of the stake holders of the educational establishment like a student, teacher, administrator, Higher management official, Managing directors or chief executives can initiate a query through an application interface which is received by the master node, master node uses the map process to assign the sub-job to the slave node which can then in turn assign the jobs to other slave nodes depending upon the architecture of the Hadoop system. The sub-jobs are executed depending upon which type of analysis needs to be done on a specific data or collection of data initiated by a query. The slave nodes finish their respective tasks and return the results to the master nodes. The process of aggregation and assembly is initiated by the master node on the varied results collected from various slave nodes. The "reduce" process is used by the master node to aggregate the results. Hadoop uses MapReduce to aggregate result collected from various slave nodes which perform operations based on the individual tasks assigned to them. The query result at the end is forwarded to the user who initiated the query from the master node as depicted in figure 1.1.

The versatility with which the above architecture can be implemented in educational establishments for providing facilities to various stake holders is exceptional as it will facilitate different users depending upon their roles and requirements in academic setup to have excellent visualization, predictive and preemptive capabilities which will certainly contribute towards establishing a better educational environment. The above architecture can be used by higher management for planning and decision making and a completely different application interface will be available to them. At the same time it will enable teachers to be wiser by providing them insights to the critical information and analytics about students. In addition the above architecture is also useful for the students to have a complete analysis of the course and modules they are taking or planning to take in the future.

IV. Big Data for Enhancing Teaching and Learning (Academic Excellence)

In educational establishments Big Data has a very good scope of implementation to enhance the teaching and learning and drive a particular organization towards academic excellence. In educational institutes lot of data that can be used for analysis and finally for the enhancement of teaching and learning is being generated every day. One of the examples which have been taken in this research paper is the use of various types of sensors in the class room to obtain huge but

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critical information about the students of a particular class room. This information can play a very vital role in planning the class room teaching, enhancing delivery, help achieving learning outcomes and very useful inputs for the student success center. The idea is to utilize electro encephalography (EEG) sensors for taking the collection of data. EEG sensor are capable of taking the brain activity, the human brain is composed up of billions of brain cells that consume electricity to interact with each other. The arrangement of millions of neurons distributing signals at once creates a massive amount of electrical movement in the brain, which can be sensed using sensitive medical apparatus to calculate electricity levels over areas of the scalp. EEG's are recordings of the electrical activity along the scalp. It evaluates voltage variations caused from ionic current flows within the neurons of the brain. A student's attentiveness while learning extensively affects his/ her learning outcomes. Teachers commonly observe students' expressions to conclude whether they are appropriately concentrating in a face to face session. However, this method is disproportionately subjective and exhausts a substantial amount of the teacher's energy.

The NeuroSky is an EEG Sensor which is available at cheap prices and measures brain's electrical activity, and can detect the level of focus. In the same manner there are few more useful sensor devices that can be utilized in the class room and are completely appropriate and wearable devices. Fig. 2 depicts the EEG sensor device that can be utilized inside the class room.

Fig. 2. Electro Encephalography Wearable Devices

Fig. 2 shows the headset EEG devices which can be easily used by the students of a particular cohort while lectures are taking place so that their level of attention can be obtained to be fed in to the big data system as depicted in the Figure 2.1. The data collected from the sensors is inputted to the Hadoop system where it is stored in the database. The analytical processing of the data is then done using variety of data analytic tools and predictive analysis can also be done using variety of tools and a desired visualization can be generated for various people like teachers, officers of student's success center etc.

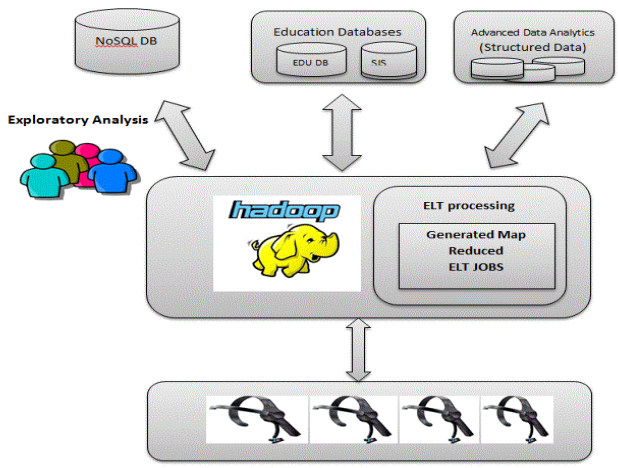

Fig. 3. Big Data Analytics

The Big Data analytics performed on the unstructured data collected from the sensors as depicted from the Fig. 3 above can prove to very beneficial for improving performance of individual students and addressing student specific problems. It is also beneficial for teachers to calculate the attention levels of students and prepare interesting session to attract more attention levels.

V. CONCLUSION

Big Data and its diverse use have already created a huge impact in many important fields and in very near future education sector is not going to be an exception. Big Data has exceptional capabilities to yield benefits to educational establishments not only in terms of providing them decision making, preventive and preemptive capabilities however making a significant contribution in enhancing teaching and learning and driving a particular educational establishment towards academic excellence. Although Big Data utilization in education sector has been exploited very less by the researchers however its impact holistically on the education system and educational establishments all over the world cannot remain mystery for too long. Big Data has a huge scope of implementation in education both in terms of enabling institutional planning/ decision support as well in achieving academic excellence by enhancing teaching and learning through non-conventional ways. Educational establishments generate a huge amount of unstructured data from diverse sources which can prove to be utmost beneficial for enhancement of teaching and learning as well as for institutional planning and decision support. Although only a few sources of data generation areas are explored in this paper however many other areas do exist in educational establishments from where unstructured huge data can be collected, that can be used for big data analytics. Need in near future would be to explore appropriate sources in educational establishments which generate huge amount of data that can be used for predictive and big data analysis to yield maximum benefit to educational establishments in terms of enhancing

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teaching and learning as well for better decision making, institutional planning, resource planning and unprecedented visualization of facts. With the exploration of more and more areas in education sector which can contribute huge data for Big Data analysis and have the scope of generating upright benefits for institutions, Big Data would surely contribute towards achieving academic excellence in educational establishments.

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REFERENCES

[1] Michael Cox and David Ellsworth, "Application-controlled demand paging for out-of-core visualization," in *Proceedings of the IEEE 8th Conference on Visualization*, 1997.

[2] TCS Big Data Study, "Manufacturing: Big Data Benefits and Challenges," Tata Consultancy Services Limited., Mumbai, Study 2014.

[3] Layton Julia. (2013, March) Money.howstuffworks.com. "http://money.howstuffworks.com/amazon1.htm" http://money.howstuffworks.com/amazon1.htm

[4] (2012, December) The Economist "Data, Data Everywhere". [Online]. “http://www.economist.com/node/15557443" http://www.economist.com/node/15557443

[5] Jay Lee, F. Wu, W. Zhao, M. Ghaffari, and L Liao, "Prognostics and health management design for rotary machinery systems—Reviews, methodology and applications"," *Mechanical Systems and Signal Processing*, vol. 1, p. 42, January 2013.

[6] Robert Johnson. (2010, July) Facebook "Scaling Facebook to 500 Million Users and Beyond". "https://www.facebook.com/notes/facebook-engineering/scaling-facebook-to-500-million-users-and-beyond/409881258919" https://www.facebook.com/notes/facebook-engineering/scaling-facebook-to-500-million-users-and-beyond/409881258919

[7] Danny Sullivan. (2015, January) Search Engine Land. [Online]. "http://searchengineland.com/google-1-trillion-searches-per-year-212940" http://searchengineland.com/google-1-trillion-searches-per-year-212940

[8] McKinsey, "Big Data: The next frontier for innovation, competition, and productivity," MCKINSEY GLOBAL INSTITUTE, 2011.

[9] P RUSSOM, "Big data analytics," TDWI Best Practices Rep, Quarterly Report 2011.

[10] Jyotsna Talreja Wassan, "Discovering Big Data Modelling for Educational World," in *IETC 2014*, New Delhi, 2014.

[11] M. S., & Kaur, A Bhullar, "Use of Data Mining in Education Sector," in *In Proceedings of the World Congress on Engineering and Computer Science*, San Francisco, 2012, pp. 24-26.

[12] Saima Mohammad, "The Use of EEG Sensors to Detect and Extract Emotions and Attention Levels from the Human Mind," in *Parsons The New School for Design* , New York, 2013.

[13] (2015) SAS Academy for Data Science. "http://www.sas.com/en_us/insights/big-data/hadoop.html"

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